Title of the Project Report

"Reliability-Aware Multimodal Emotion Recognition with Adaptive Fusion, Conflict Handling and Missing Modality Robustness"

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CERTIFICATE

This is to certify that the work contained in this report entitled "Title of the Project Report" is submitted by the Ms. Shraddha Verma to the Department of Computer Science and Engineering, National Institute of Technology Goa, for the partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering.

She has carried out their work under my supervision. This work has not been submitted elsewhere for the award of any other degree or diploma.

The project work in our opinion, has reached the standards fulfilling the requirements for the degree of Bachelor of Technology in Computer Science and Engineering in accordance with the regulations of the Institute.

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ABSTRACT

Emotion recognition plays a crucial role in human-computer interaction, healthcare, education, customer service, and intelligent assistive systems. Traditional emotion recognition systems primarily rely on a single modality such as text, speech, or facial expressions. However, unimodal approaches often suffer from reduced accuracy and poor robustness when the available information is noisy, incomplete, or ambiguous. To address these limitations, this project presents a Reliability-Aware Multimodal Emotion Recognition framework that integrates information from text, audio, and visual modalities for improved emotion classification.

The proposed system employs modality-specific feature extraction techniques to analyze emotional cues from multiple sources. Textual information is processed using transformer-based language models, audio signals are analyzed through deep learning-based speech emotion recognition techniques, and facial expressions are examined using computer vision methods. The extracted features are combined through an adaptive fusion mechanism that dynamically assigns weights to individual modalities based on their reliability and confidence scores.

A key contribution of this work is the incorporation of a Reliability Scoring Module that evaluates the trustworthiness of each modality before fusion. The framework also includes conflict handling and missing modality robustness mechanisms, enabling the system to maintain stable performance even when one or more modalities provide unreliable or unavailable information. This approach improves the overall robustness and interpretability of multimodal emotion recognition systems.

The proposed framework is evaluated using publicly available multimodal emotion datasets, including MELD and RAVDESS. Experimental observations demonstrate that reliability-aware adaptive fusion provides more consistent and accurate emotion predictions compared to conventional equal-weight fusion approaches. The developed system offers a practical and scalable solution for real-world emotion-aware applications and highlights the potential of uncertainty-aware multimodal learning in affective computing.

Keywords: Multimodal Emotion Recognition, Emotion Classification, Reliability Scoring, Adaptive Fusion, Deep Learning, BERT, Speech Emotion Recognition, Computer Vision, Affective Computing.

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CHAPTER 1
INTRODUCTION

1.1 Background

Human emotions play a significant role in communication, decision-making, social interactions, and behavior understanding. The ability of machines to recognize and interpret human emotions has become an important research area in Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), and Computer Vision (CV). Emotion Recognition aims to automatically identify a person's emotional state from various forms of data such as text, speech, facial expressions, physiological signals, and gestures.

Traditional emotion recognition systems are primarily based on a single source of information, commonly referred to as unimodal emotion recognition. For example, textual emotion recognition analyzes the sentiment and emotional content of written text, speech emotion recognition utilizes vocal characteristics such as pitch and intensity, while facial emotion recognition relies on visual cues obtained from facial expressions. Although these approaches have achieved considerable success, their performance is often limited by noise, ambiguity, missing information, and variations in real-world environments.

To overcome these limitations, researchers have increasingly focused on Multimodal Emotion Recognition (MER), which combines information from multiple modalities such as text, audio, and visual data. By integrating complementary emotional cues from different sources, multimodal systems can achieve improved accuracy, robustness, and reliability compared to unimodal approaches. Multimodal emotion recognition has found applications in intelligent virtual assistants, mental health monitoring, e-learning platforms, customer service systems, healthcare, human-computer interaction, and social robotics.

Despite recent advancements, several challenges continue to hinder the practical deployment of multimodal emotion recognition systems. These challenges include modality conflicts, unreliable predictions from individual modalities, missing modality situations, noisy data, and difficulties in effectively fusing heterogeneous information. In many real-world scenarios, one modality may provide misleading or incomplete emotional cues, resulting in reduced overall system performance.

To address these challenges, this project proposes a Reliability-Aware Multimodal Emotion Recognition framework that incorporates adaptive fusion, confidence estimation, conflict handling, and missing modality robustness. The proposed approach dynamically evaluates the reliability of each modality before combining their predictions, enabling more accurate and trustworthy emotion recognition. By leveraging reliability-aware decision-making, the system aims to improve performance under realistic operating conditions while enhancing the interpretability and robustness of multimodal emotion recognition systems.

1.2 Motivation

The increasing adoption of intelligent systems in everyday life has created a growing demand for machines that can understand and respond to human emotions effectively. Applications such as virtual assistants, online learning platforms, healthcare monitoring systems, customer support services, and social robots require accurate emotion recognition capabilities to provide personalized and human-centric interactions. However, existing unimodal emotion recognition systems often struggle to perform reliably in real-world environments due to incomplete, noisy, or ambiguous information.

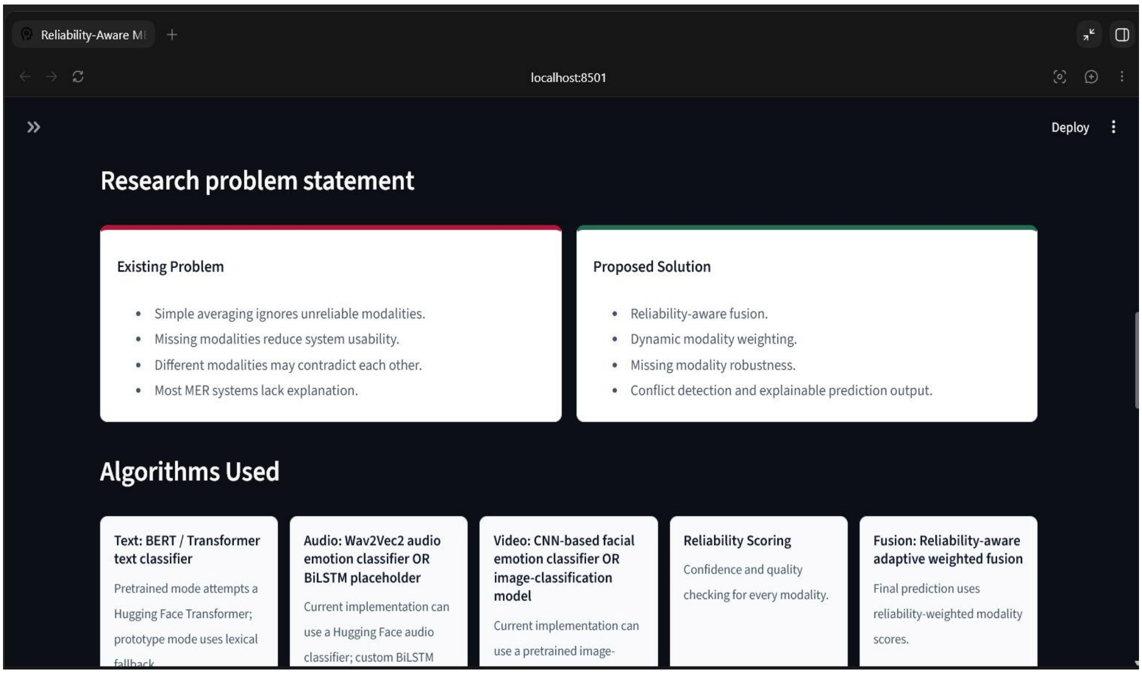
Although multimodal emotion recognition has demonstrated significant improvements over unimodal approaches, many existing systems treat all modalities equally during the fusion process. In practice, the reliability of different modalities may vary considerably depending on the context. For instance, facial expressions may be partially occluded, audio recordings may contain background noise, or textual information may lack sufficient emotional cues. Assigning equal importance to all modalities in such situations can lead to inaccurate predictions and reduced system performance.

This limitation motivated the development of a reliability-aware framework that can dynamically evaluate the trustworthiness of each modality before combining their outputs. By incorporating confidence estimation, adaptive fusion, and conflict handling mechanisms, the proposed system aims to make more informed decisions and maintain robust performance even when one or more modalities become unreliable or unavailable.

The motivation behind this work is to contribute towards the development of practical and dependable emotion recognition systems capable of operating effectively in real-world scenarios. The proposed approach seeks to improve prediction accuracy, enhance system robustness, and provide a foundation for future research in uncertainty-aware multimodal affective computing.

1.3 Problem Statement

Multimodal Emotion Recognition (MER) systems aim to improve emotion classification performance by combining information from multiple modalities such as text, speech, and facial expressions. Although recent advances in deep learning have significantly enhanced the effectiveness of multimodal approaches, several challenges still limit their reliability and practical deployment in real-world applications.



One of the major challenges is the varying reliability of different modalities under different conditions. A modality that provides strong emotional cues in one situation may become unreliable in another due to noise, missing data, poor quality inputs, or ambiguous expressions. Most existing multimodal emotion recognition systems employ fixed or equal-weight fusion strategies, which assume that all modalities contribute equally to the final prediction. Such assumptions often result in suboptimal performance when one or more modalities provide misleading or conflicting information.

Another challenge is the presence of modality conflicts, where different modalities predict different emotions for the same instance. Existing fusion methods frequently lack effective mechanisms to resolve such conflicts, leading to uncertainty in the final decision. Furthermore, real-world applications often encounter missing modality scenarios caused by sensor failures, unavailable inputs, or incomplete data, which can significantly degrade system performance.

Therefore, there is a need for a robust multimodal emotion recognition framework capable of dynamically assessing the reliability of individual modalities, handling conflicting predictions, and maintaining stable performance in the presence of missing or noisy data. The problem addressed in this work is the development of a Reliability-Aware Multimodal Emotion Recognition system that utilizes adaptive fusion and confidence-based decision making to improve the accuracy, robustness, and trustworthiness of emotion recognition in real-world environments.

1.4 Objectives

- The primary objective of this project is to develop a reliable and robust multimodal emotion recognition framework capable of accurately identifying human emotions from multiple data modalities. The specific objectives of this work are as follows:

- To study existing emotion recognition techniques based on text, audio, and visual modalities.

- To develop a multimodal emotion recognition system that integrates information from text, speech, and facial expressions.

- To implement modality-specific feature extraction techniques for capturing emotional cues from different sources.

- To design a Reliability Scoring Module for evaluating the confidence and trustworthiness of individual modalities.

- To develop an adaptive fusion mechanism that dynamically assigns weights to modalities based on their reliability scores.

- To handle conflicting predictions generated by different modalities through confidence-based decision making.

- To improve system robustness in scenarios where one or more modalities are missing or unreliable.

- To evaluate the effectiveness of the proposed framework using benchmark multimodal emotion datasets such as MELD and RAVDESS.

- To compare the performance of the proposed reliability-aware fusion approach with conventional fusion techniques.

- To develop a practical and scalable framework suitable for real-world emotion-aware applications.

1.5 Proposed Contributions

The major contributions of the proposed work are summarized as follows:

- A Reliability-Aware Multimodal Emotion Recognition framework is proposed to improve the accuracy and robustness of emotion classification.

- A Reliability Scoring Module is introduced to estimate the confidence and trustworthiness of individual modalities before fusion.

- An adaptive weighted fusion mechanism is developed to dynamically assign importance to text, audio, and visual modalities based on their reliability.

- A conflict handling strategy is incorporated to resolve situations where different modalities predict conflicting emotions.

- A missing modality robustness mechanism is implemented to ensure stable system performance even when one or more modalities are unavailable.

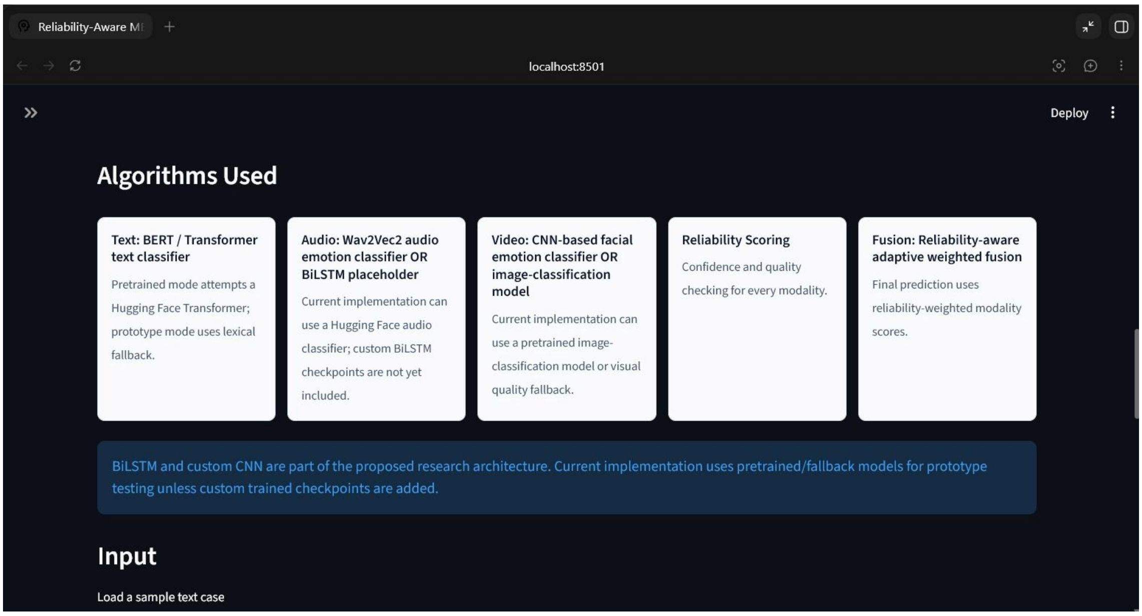
- The proposed framework combines textual, speech, and facial emotional cues to achieve more comprehensive emotion understanding.

- Experimental evaluation is performed using benchmark multimodal emotion datasets such as MELD and RAVDESS.

- Comparative analysis is conducted to demonstrate the advantages of reliability-aware fusion over conventional fusion approaches.

- The developed system provides improved interpretability, robustness, and practical applicability for real-world emotion-aware applications.

- The proposed framework establishes a foundation for future research in uncertainty-aware and reliability-driven multimodal affective computing systems.



1.6 Organization of the Report

This report is organized into five chapters. Chapter 1 presents the introduction, motivation, problem statement, objectives, proposed contributions, and overall scope of the work. Chapter 2 provides a review of existing literature and recent advancements in the field of multimodal emotion recognition. Chapter 3 describes the proposed Reliability-Aware Multimodal Emotion Recognition framework, including the system architecture, reliability scoring mechanism, adaptive fusion strategy, and conflict handling approach. Chapter 4 presents the experimental setup, datasets used, implementation details, performance evaluation, and results obtained from the proposed system. Finally, Chapter 5 summarizes the major findings of the work, discusses its limitations, and outlines potential directions for future research.

CHAPTER 2
LITERATURE REVIEW

2.1 Background of Existing Research

Emotion recognition has emerged as an important research area in Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), and Computer Vision (CV). The primary objective of emotion recognition systems is to automatically identify and classify human emotional states from various forms of data such as text, speech, facial expressions, physiological signals, and behavioral patterns. Accurate emotion recognition enables intelligent systems to better understand human behavior and provide more personalized, context-aware, and effective interactions.

With the rapid growth of human-computer interaction technologies, emotion recognition has gained significant attention in applications such as healthcare monitoring, virtual assistants, intelligent tutoring systems, customer service, social robotics, entertainment, and mental health assessment. Traditional approaches mainly focused on analyzing a single source of information; however, human emotions are naturally expressed through multiple channels simultaneously. Therefore, relying on a single modality often leads to incomplete understanding and reduced recognition performance.

Recent advancements in deep learning and multimodal learning have encouraged researchers to combine information from multiple modalities, including text, audio, and visual data, to improve emotion recognition accuracy. Multimodal Emotion Recognition (MER) has demonstrated superior performance by leveraging complementary emotional cues from different sources. Despite these advancements, several challenges such as modality conflicts, missing data, unreliable predictions, and fusion complexity continue to affect the effectiveness of existing systems.

This chapter presents a review of existing research related to emotion recognition, multimodal learning, fusion techniques, and the challenges associated with current Multimodal Emotion Recognition systems. The literature review provides the foundation for identifying research gaps and motivating the development of the proposed Reliability-Aware Multimodal Emotion Recognition framework.

2.2 Emotion Recognition

Emotion Recognition refers to the process of identifying and classifying human emotional states using computational techniques. Human emotions are expressed through various channels, including spoken language, facial expressions, body gestures, physiological responses, and written text. The objective of emotion recognition systems is to automatically analyze these emotional cues and determine the underlying emotional state of an individual.

The field of emotion recognition is closely associated with Affective Computing, a branch of Artificial Intelligence that focuses on enabling machines to recognize, interpret, and respond to human emotions. Effective emotion recognition allows intelligent systems to interact with users in a more natural, personalized, and human-centric manner. As a result, emotion recognition has become a key component in modern applications such as virtual assistants, healthcare monitoring systems, online education platforms, customer support systems, social robots, and smart surveillance systems.

Emotion recognition systems generally classify emotions into predefined categories such as happiness, sadness, anger, fear, surprise, disgust, and neutral state. Some approaches also represent emotions using dimensional models, where emotional states are characterized by continuous values such as valence, arousal, and dominance. Both categorical and dimensional approaches are widely used in affective computing research depending on the application requirements.

Traditionally, emotion recognition relied on handcrafted feature extraction techniques and conventional machine learning algorithms. However, recent advancements in deep learning have significantly improved recognition performance by enabling automatic feature learning from large-scale datasets. Techniques based on Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, Transformers, and Attention Mechanisms have achieved remarkable success in extracting complex emotional patterns from different modalities.

Despite these advancements, accurately recognizing emotions remains a challenging task due to variations in human behavior, cultural differences, noisy inputs, ambiguous emotional expressions, and contextual dependencies. These challenges have motivated researchers to explore multimodal approaches that combine information from multiple sources to achieve more reliable and robust emotion recognition.

2.3 Unimodal Emotion Recognition

Unimodal Emotion Recognition refers to the process of identifying human emotions using a single source of information. Depending on the type of input data, unimodal systems may analyze textual content, speech signals, facial expressions, physiological signals, or body gestures to determine the emotional state of an individual. Due to their relatively simple architecture and lower computational complexity, unimodal approaches have been extensively studied and widely adopted in emotion recognition research.

2.3.1 Text-Based Emotion Recognition

Text-based emotion recognition focuses on identifying emotions from written language such as social media posts, chat messages, reviews, emails, and conversational transcripts. Traditional approaches relied on sentiment lexicons and handcrafted linguistic features, while modern systems utilize deep learning and transformer-based models such as BERT, RoBERTa, and GPT-based architectures for contextual understanding. Text-based approaches are effective in capturing semantic and contextual information; however, they often struggle when emotional cues are implicit or when contextual information is insufficient.

2.3.2 Speech-Based Emotion Recognition

Speech-based emotion recognition aims to detect emotions from vocal characteristics present in audio signals. Emotional information can be extracted from various acoustic features such as pitch, intensity, speech rate, energy, Mel-Frequency Cepstral Coefficients (MFCCs), and spectral characteristics. Deep learning techniques including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) networks have significantly improved speech emotion recognition performance. However, environmental noise, speaker variability, and recording quality can negatively affect system accuracy.

2.3.3 Facial Emotion Recognition

Facial Emotion Recognition (FER) analyzes facial expressions to identify emotional states. Human faces contain rich emotional information through movements of facial muscles, eye regions, eyebrows, and mouth positions. Traditional computer vision techniques used handcrafted features, whereas modern systems employ deep learning models such as CNNs and Vision Transformers (ViTs) for automatic feature extraction. Facial emotion recognition has achieved impressive performance in controlled environments but may face challenges due to illumination variations, occlusions, pose changes, and image quality issues.

2.3.4 Limitations of Unimodal Approaches

Although unimodal emotion recognition systems have achieved considerable success, they suffer from several limitations. Since emotions are complex and multidimensional, relying on a single modality often provides incomplete information about an individual's emotional state. Text may fail to capture vocal emotions, speech may not reflect facial expressions, and visual cues alone may not provide contextual understanding. Additionally, noise, ambiguity, missing information, and environmental factors can significantly impact performance. These limitations have motivated the development of Multimodal Emotion Recognition systems that combine information from multiple modalities to improve accuracy, robustness, and reliability.

2.4 Multimodal Emotion Recognition

Multimodal Emotion Recognition (MER) is an advanced approach that aims to identify human emotions by simultaneously analyzing multiple sources of information, such as text, speech, facial expressions, physiological signals, and body gestures. Unlike unimodal systems, which rely on a single modality, multimodal approaches leverage complementary emotional cues from different modalities to achieve more accurate and reliable emotion classification.

Human emotions are naturally expressed through multiple channels. For example, during a conversation, a person may express emotions through spoken words, vocal tone, facial expressions, and body language simultaneously. A unimodal system may fail to capture the complete emotional context if one modality is ambiguous or noisy. By integrating information from multiple modalities, multimodal emotion recognition systems can provide a more comprehensive understanding of emotional states and significantly improve recognition performance.

The general architecture of a multimodal emotion recognition system consists of three major stages: data acquisition, feature extraction, and multimodal fusion. In the data acquisition stage, information is collected from different modalities such as text transcripts, audio recordings, and facial images or videos. Feature extraction techniques are then applied independently to each modality to obtain meaningful emotional representations. Finally, the extracted features are combined using a fusion mechanism to generate the final emotion prediction.

Recent advancements in deep learning have greatly enhanced the performance of multimodal emotion recognition systems. Transformer-based language models such as BERT are commonly used for textual feature extraction, while Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) networks are frequently employed for speech and visual emotion analysis. Attention mechanisms and multimodal transformer architectures have further improved the ability of models to capture relationships between different modalities.

Despite achieving higher accuracy than unimodal approaches, multimodal emotion recognition systems still face several challenges. Differences in modality quality, synchronization issues, missing modality scenarios, conflicting predictions, and fusion complexity can negatively affect overall system performance. Consequently, researchers have increasingly focused on developing adaptive and reliability-aware fusion techniques capable of dynamically handling uncertain and inconsistent multimodal information.

Due to its superior ability to capture complex emotional patterns, Multimodal Emotion Recognition has become a rapidly growing research area with applications in healthcare, intelligent tutoring systems, human-computer interaction, social robotics, customer service, and mental health monitoring. The continued advancement of multimodal learning techniques is expected to play a crucial role in the development of more intelligent and emotionally aware systems.

2.5 Existing Fusion Techniques

Fusion is a critical component of Multimodal Emotion Recognition systems, as it determines how information from different modalities is combined to generate the final emotion prediction. The effectiveness of the fusion strategy directly influences the overall performance, robustness, and reliability of the system. Over the years, researchers have proposed various fusion techniques to integrate emotional information obtained from text, speech, and visual modalities.

2.5.1 Early Fusion

Early Fusion, also known as feature-level fusion, combines features extracted from different modalities before classification. In this approach, modality-specific features are concatenated into a single feature vector, which is then provided as input to a machine learning or deep learning model. Early fusion enables the model to learn relationships between modalities during the training process and often captures cross-modal interactions effectively.

However, early fusion faces several challenges. Different modalities may have varying feature dimensions, data distributions, and temporal characteristics, making feature alignment difficult. Furthermore, missing data from any modality can significantly affect the performance of the entire system.

2.5.2 Late Fusion

Late Fusion, also referred to as decision-level fusion, performs independent emotion prediction for each modality and combines the resulting predictions at the final stage. Common techniques include majority voting, weighted averaging, and confidence-based aggregation. Late fusion offers greater flexibility and can handle missing modalities more effectively since each modality is processed independently.

Despite its advantages, late fusion may fail to fully exploit the relationships and dependencies that exist between different modalities, potentially limiting overall performance.

2.5.3 Hybrid Fusion

Hybrid Fusion combines the strengths of both early and late fusion approaches. It integrates information at multiple stages of the processing pipeline, allowing the system to capture both feature-level and decision-level relationships. Hybrid fusion strategies have demonstrated improved performance in various multimodal learning applications; however, they often require more complex architectures and higher computational resources.

2.5.4 Attention-Based Fusion

Recent advances in deep learning have led to the development of attention-based fusion mechanisms. Attention models dynamically learn the importance of different modalities and assign higher weights to more informative inputs. These techniques enable the system to focus on relevant emotional cues while reducing the influence of noisy or less informative modalities. Attention-based fusion has achieved promising results in multimodal emotion recognition and other multimodal learning tasks.

2.5.5 Limitations of Existing Fusion Techniques

Although existing fusion methods have improved the performance of multimodal emotion recognition systems, several limitations remain. Most traditional fusion approaches assume that all modalities contribute equally or rely on fixed weighting schemes that do not adapt to changing conditions. In real-world scenarios, modality reliability can vary significantly due to noise, occlusions, poor audio quality, ambiguous text, or missing inputs. Consequently, fixed fusion strategies may produce inaccurate predictions when unreliable modalities influence the final decision.

These limitations highlight the need for reliability-aware fusion mechanisms that can dynamically evaluate the confidence of individual modalities and adapt the fusion process accordingly. Such approaches can improve robustness, handle conflicting predictions more effectively, and maintain stable performance in realistic operating environments.

2.6 Challenges in Existing MER Systems

Despite significant advancements in Multimodal Emotion Recognition (MER), several challenges continue to limit the effectiveness and real-world applicability of existing systems. The complexity of human emotions, variability in data quality, and differences among modalities make accurate emotion recognition a difficult task. These challenges often affect the robustness, reliability, and generalization capability of multimodal emotion recognition models.

2.6.1 Modality Reliability Variations

Different modalities do not always contribute equally to emotion recognition. In some situations, facial expressions may provide strong emotional cues, while in others, speech or textual information may be more informative. Most existing systems fail to account for these variations and often assume equal importance for all modalities, resulting in suboptimal performance.

2.6.2 Modality Conflicts

A common challenge in multimodal emotion recognition arises when different modalities predict different emotions for the same instance. For example, a person's facial expression may indicate sadness while the spoken text suggests happiness. Existing fusion methods often struggle to effectively resolve such conflicts, leading to uncertainty in the final prediction.

2.6.3 Missing Modality Problem

Real-world applications frequently encounter situations where one or more modalities are unavailable due to sensor failures, poor data quality, network issues, or incomplete recordings. Many multimodal systems are highly dependent on the availability of all modalities and experience significant performance degradation when information from one modality is missing.

2.6.4 Noise and Data Quality Issues

Audio recordings may contain background noise, facial images may suffer from occlusions or poor lighting conditions, and textual inputs may contain ambiguous or incomplete information. These factors can negatively affect feature extraction and emotion classification, reducing the overall reliability of the system.

2.6.5 Fusion Complexity

Combining heterogeneous information from multiple modalities remains a challenging task. Differences in feature dimensions, temporal alignment, and semantic representation make it difficult to design efficient fusion mechanisms. Improper fusion strategies can result in information loss and reduced model performance.

2.6.6 Generalization and Real-World Deployment

Many existing models achieve high performance on benchmark datasets but fail to generalize effectively to unseen environments.

Variations in language, culture, recording conditions, and user behavior often lead to reduced performance in practical applications. Developing robust systems that perform consistently across diverse real-world scenarios remains a major research challenge.

2.6.7 Lack of Interpretability

Deep learning-based multimodal systems often operate as black-box models, making it difficult to understand the reasoning behind their predictions. The lack of transparency reduces user trust and limits their adoption in critical applications such as healthcare, education, and mental health monitoring.

The presence of these challenges highlights the need for more robust, adaptive, and reliability-aware multimodal emotion recognition frameworks. Addressing these limitations can significantly improve the accuracy, trustworthiness, and practical applicability of future emotion recognition systems.

2.7 Research Gap

The literature review indicates that significant progress has been made in the field of Multimodal Emotion Recognition through the use of deep learning architectures, transformer-based models, and advanced fusion techniques. Existing studies have demonstrated that combining information from multiple modalities generally produces better performance than unimodal approaches. However, several important limitations still remain insufficiently addressed.

Most existing multimodal emotion recognition systems employ fixed fusion strategies or assume equal contribution from all modalities during the decision-making process. In practical scenarios, the quality and reliability of different modalities may vary significantly due to noise, incomplete data, environmental conditions, or ambiguous emotional expressions. As a result, treating all modalities equally can lead to inaccurate predictions and reduced system robustness.

Another important limitation is the lack of effective mechanisms for handling modality conflicts. When different modalities predict different emotional states, many existing systems fail to determine which modality should be trusted more. Similarly, missing modality scenarios remain a major challenge, as numerous multimodal models are designed under the assumption that all modalities will always be available during inference.

Furthermore, although attention-based fusion techniques have improved multimodal learning performance, many existing approaches do not explicitly quantify modality reliability or confidence before combining predictions. Consequently, unreliable modalities may still influence the final decision, reducing the trustworthiness and interpretability of the system.

Based on the identified limitations, a clear research gap exists in the development of emotion recognition systems that can dynamically evaluate modality reliability, resolve conflicting predictions, and maintain robust performance under missing modality conditions. Therefore, this work proposes a Reliability-Aware Multimodal Emotion Recognition framework that incorporates confidence estimation, adaptive weighted fusion, conflict handling, and missing modality robustness mechanisms. The proposed approach aims to address the shortcomings of existing systems and improve the accuracy, reliability, and practical applicability of multimodal emotion recognition in real-world environments.

2.8 Chapter Summary

This chapter presented a comprehensive review of emotion recognition and multimodal emotion recognition systems. The discussion covered the fundamental concepts of emotion recognition, the characteristics of unimodal and multimodal approaches, and the various fusion techniques employed in existing research. The strengths and limitations of early fusion, late fusion, hybrid fusion, and attention-based fusion methods were examined. Furthermore, several challenges associated with current Multimodal Emotion Recognition systems, including modality reliability variations, modality conflicts, missing modality scenarios, noise, fusion complexity, generalization issues, and lack of interpretability, were discussed.

Based on the literature review, a significant research gap was identified in the development of emotion recognition systems capable of dynamically evaluating modality reliability and effectively handling uncertain or conflicting information. To address these limitations, the proposed work introduces a Reliability-Aware Multimodal Emotion Recognition framework incorporating adaptive fusion, confidence estimation, conflict handling, and missing modality robustness mechanisms. The next chapter presents the proposed methodology, system architecture, and implementation details of the proposed framework.

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CHAPTER 3
PROPOSED METHODOLOGY

3.1 Overview of the Proposed Framework

The proposed Reliability-Aware Multimodal Emotion Recognition framework is designed to improve the accuracy, robustness, and reliability of emotion classification by effectively integrating information obtained from multiple modalities, including text, audio, and visual data. The framework addresses several limitations of existing multimodal emotion recognition systems, particularly those related to modality conflicts, unreliable predictions, noisy inputs, and missing modality scenarios.

The proposed system follows a multi-stage processing pipeline consisting of data acquisition, preprocessing, feature extraction, reliability estimation, adaptive fusion, and final emotion classification. Initially, emotional information is collected from text, speech, and facial expression modalities. Each modality is then processed independently using specialized preprocessing and feature extraction techniques to obtain meaningful emotional representations.

After feature extraction, a Reliability Scoring Module is employed to estimate the confidence and trustworthiness of each modality. The reliability score reflects the quality and consistency of the information provided by a modality and serves as an indicator of its contribution to the final decision-making process. Modalities with higher confidence are assigned greater importance, while unreliable modalities receive lower influence during fusion.

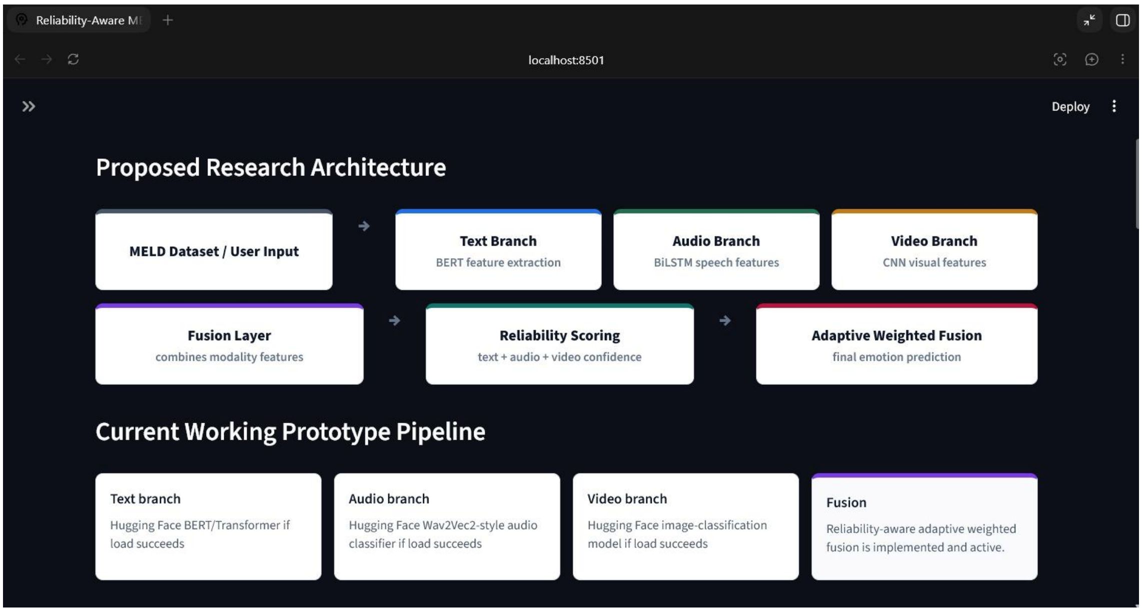
The framework further incorporates an Adaptive Fusion Mechanism that dynamically combines modality-specific predictions according to their reliability scores. Unlike conventional fusion approaches that utilize fixed or equal weights, the proposed method continuously adjusts modality contributions based on their estimated reliability. This enables the system to make more informed and robust decisions under varying operating conditions.

To further enhance robustness, the framework includes a Conflict Handling Strategy capable of resolving situations where different modalities produce contradictory emotion predictions. Additionally, a Missing Modality Robustness mechanism ensures stable performance even when one or more modalities are unavailable or contain incomplete information.

The overall objective of the proposed framework is to develop a practical and reliable multimodal emotion recognition system capable of operating effectively in real-world environments. By integrating reliability-aware decision making, adaptive fusion, and uncertainty handling, the proposed approach aims to improve emotion classification performance while enhancing interpretability and trustworthiness.

3.2 System Architecture

The proposed Reliability-Aware Multimodal Emotion Recognition framework follows a structured architecture that integrates information from text, audio, and visual modalities to perform robust emotion classification. The architecture is designed to effectively process multimodal data, evaluate modality reliability, and generate accurate emotion predictions through adaptive fusion.



The overall system consists of six major components: Data Acquisition, Data Preprocessing, Feature Extraction, Reliability Scoring, Adaptive Fusion, and Emotion Classification. Each component performs a specific function within the emotion recognition pipeline and contributes to the overall effectiveness of the framework.

Initially, multimodal emotional data are collected from benchmark datasets such as MELD and RAVDESS. The acquired data consist of textual utterances, speech recordings, and facial expression information. Since raw data may contain noise, inconsistencies, and irrelevant information, preprocessing techniques are applied separately to each modality to improve data quality and prepare the inputs for feature extraction.

Following preprocessing, modality-specific feature extraction methods are employed. Textual features are extracted using transformer-based language models capable of capturing semantic and contextual information. Audio features are obtained from speech signals using acoustic feature extraction techniques, while visual features are extracted from facial images or video frames using deep learning-based computer vision models. These features provide rich emotional representations from their respective modalities.

The extracted modality-specific features are then passed to the Reliability Scoring Module. This module estimates the confidence and trustworthiness of each modality by analyzing prediction consistency and feature quality. The resulting reliability scores indicate the relative importance of each modality during the fusion process.

Based on the reliability scores, an Adaptive Fusion Mechanism dynamically assigns weights to modality-specific predictions and combines them to generate a unified emotional representation. The framework further incorporates conflict handling and missing modality management strategies to improve robustness under uncertain conditions.

Finally, the fused representation is processed by the emotion classification module, which predicts the final emotional category. The overall architecture enables the system to adapt to varying modality quality, reduce the impact of unreliable inputs, and provide more accurate and trustworthy emotion recognition results compared to conventional fusion-based approaches.

System Flow:

Text Input + Audio Input + Video Input

→ Data Preprocessing

→ Feature Extraction

→ Reliability Scoring Module

→ Adaptive Weighted Fusion

→ Conflict Handling & Missing Modality Processing

→ Final Emotion Classification

3.3 Dataset Description

The performance of any emotion recognition system largely depends on the quality and diversity of the datasets used for training and evaluation. In this work, publicly available benchmark datasets are utilized to develop and assess the proposed Reliability-Aware Multimodal Emotion Recognition framework. The selected datasets provide emotional information from multiple modalities and are widely used in multimodal emotion recognition research.

3.3.1 MELD Dataset

The Multimodal EmotionLines Dataset (MELD) is one of the most widely used benchmark datasets for multimodal emotion recognition in conversations. The dataset is derived from the television series *Friends* and contains conversational utterances annotated with emotion labels. MELD provides textual transcripts, speech recordings, and visual information, making it suitable for multimodal learning applications.

The dataset contains multiple emotion categories, including happiness, sadness, anger, fear, surprise, disgust, and neutral emotions. Since

MELD captures real conversational interactions involving multiple speakers and varying emotional expressions, it serves as an effective benchmark for evaluating multimodal emotion recognition systems under realistic conditions.

3.3.2 RAVDESS Dataset

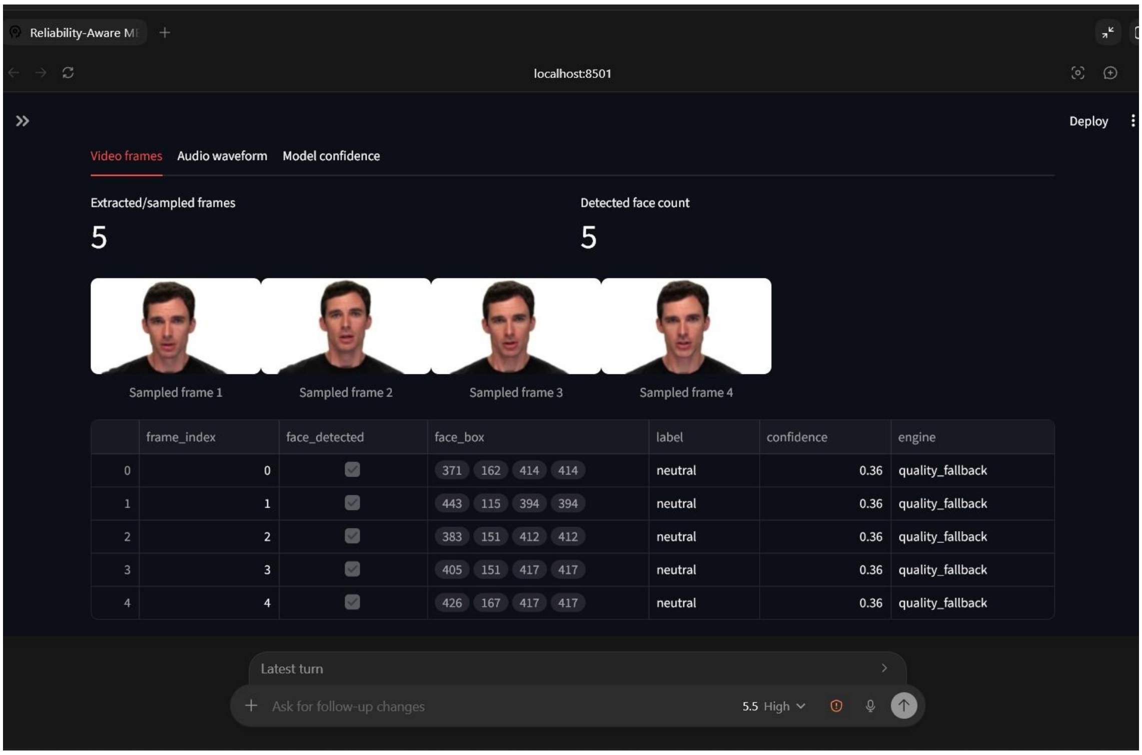
The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) is a widely used dataset for speech and facial emotion recognition research. It contains audio recordings and video clips of professional actors expressing different emotions through speech and facial expressions.

The dataset includes emotional categories such as neutral, calm, happy, sad, angry, fearful, disgust, and surprised. RAVDESS provides high-quality audio and visual recordings, making it suitable for studying emotional patterns across speech and facial modalities. The controlled recording environment of the dataset helps ensure consistency and reliability during model evaluation.

3.3.3 Dataset Utilization in the Proposed Framework

In the proposed system, the datasets are used to extract emotional information from text, audio, and visual modalities. Textual utterances are processed to capture semantic and contextual emotional cues, audio recordings are analyzed for speech-based emotional characteristics, and visual data are utilized to identify facial expressions associated with different emotions.

The combination of MELD and RAVDESS enables the proposed framework to learn diverse emotional patterns and evaluate the effectiveness of reliability-aware adaptive fusion under various multimodal scenarios. The use of benchmark datasets also facilitates fair comparison with existing emotion recognition approaches reported in the literature.



3.3.4 Summary of Datasets

<u>Summary of Datasets Used in the Proposed Framework</u>

Datasets	Modalities Available	Emotion Categories	Primary Usage
MELD (Multimodal EmotionLines Dataset)	Text, Audio, Video	Happiness, Sadness, Anger, Fear, Surprise, Disgust, Neutral	Multimodal Conversational Emotion Recognition
RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song)	Audio, Video	Neutral, Calm, Happy, Sad, Angry, Fearful, Disgust, Surprised	Speech and Facial Emotion Recognition

The selected datasets provide comprehensive multimodal emotional information and serve as a strong foundation for developing and evaluating the proposed Reliability-Aware Multimodal Emotion Recognition framework.

3.4 Data Preprocessing

Data preprocessing is a crucial step in the proposed Reliability-Aware Multimodal Emotion Recognition framework. Raw multimodal data often contain noise, inconsistencies, redundant information, and variations in format that can negatively affect model performance. Therefore, preprocessing techniques are applied separately to each modality to improve data quality and ensure effective feature extraction.

3.4.1 Text Preprocessing

Textual data obtained from conversational transcripts are cleaned and normalized before feature extraction. The preprocessing steps include:

* Removal of unnecessary symbols, special characters, and punctuation marks.
* Conversion of text into a standardized format.
* Tokenization of sentences into individual words or sub-word units.
* Elimination of irrelevant spaces and formatting inconsistencies.
* Preparation of text inputs for transformer-based language models.

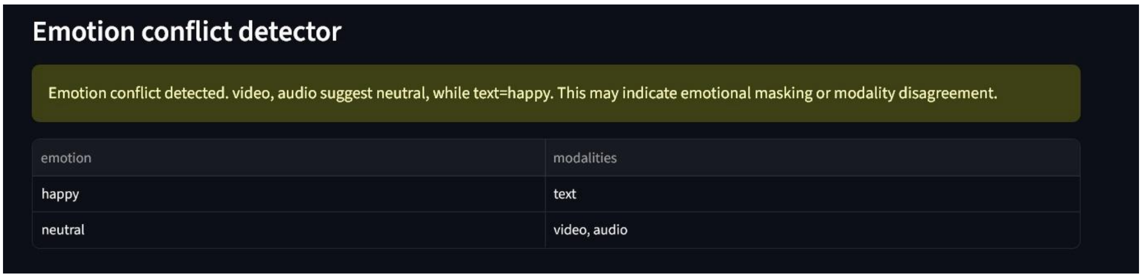
These preprocessing operations help preserve meaningful semantic information while reducing noise in the textual modality.

3.4.2 Audio Preprocessing

Speech recordings often contain background noise and variations in recording quality. Therefore, several preprocessing techniques are applied to improve the quality of audio signals:

* Noise reduction and signal enhancement.
* Normalization of audio amplitude levels.
* Removal of silent and low-information segments.
* Conversion of audio signals into a consistent sampling format.
* Preparation of speech data for acoustic feature extraction.

These steps improve the quality of emotional information extracted from speech signals.



3.4.3 Visual Preprocessing

Visual data obtained from facial images and video frames are processed to ensure consistency and improve facial feature extraction. The preprocessing steps include:

* Face detection and localization.
* Face alignment and cropping.
* Image resizing to a fixed input dimension.
* Pixel value normalization.
* Removal of irrelevant background information.

These operations enable the visual model to focus on facial expressions that carry emotional information.

3.4.4 Preprocessed Data Output

After preprocessing, the textual, audio, and visual data become standardized and suitable for feature extraction. The resulting processed inputs are forwarded to the modality-specific feature extraction modules, where meaningful emotional representations are generated for subsequent reliability estimation and adaptive fusion.

The preprocessing stage plays a vital role in improving data quality, reducing noise, and enhancing the overall performance of the proposed multimodal emotion recognition framework.

3.5 Reliability-Aware Adaptive Fusion

The Reliability-Aware Adaptive Fusion module is the core component of the proposed Multimodal Emotion Recognition framework. Its primary objective is to combine emotional information obtained from text, audio, and visual modalities in a manner that reflects the reliability of each modality. Unlike conventional fusion methods that assign equal importance to all modalities, the proposed approach dynamically adjusts modality contributions according to their confidence and quality.

Initially, modality-specific features are extracted from textual, audio, and visual inputs using appropriate feature extraction techniques. Each modality independently generates emotion prediction probabilities corresponding to different emotional categories. These predictions are then evaluated by a reliability assessment mechanism that estimates the confidence of each modality.

Based on the computed reliability scores, adaptive weights are assigned to the modalities. Modalities exhibiting higher confidence receive greater importance during fusion, while less reliable modalities contribute proportionally less to the final decision. This reliability-aware weighting strategy enables the system to reduce the influence of noisy, ambiguous, or low-quality inputs.

The proposed fusion mechanism also addresses modality conflicts by prioritizing predictions from more reliable modalities when contradictory emotional cues are observed. Furthermore, in situations where one or more modalities are unavailable, the framework automatically redistributes weights among the remaining modalities, ensuring stable system performance under missing modality conditions.

By integrating reliability estimation, adaptive weighting, conflict resolution, and missing modality handling within a unified fusion framework, the proposed approach enhances the robustness, accuracy, and practical applicability of multimodal emotion recognition. The resulting fused representation is subsequently utilized to generate the final emotion prediction.

3.6 System Workflow

The proposed Reliability-Aware Multimodal Emotion Recognition framework follows a systematic workflow that integrates textual, audio, and visual information to generate accurate and robust emotion predictions. The workflow consists of multiple stages, beginning with multimodal data acquisition and ending with the final emotion classification. Each stage contributes to improving the reliability and effectiveness of the overall system.

Step 1: Multimodal Data Acquisition

The process begins with the collection of multimodal emotional data from benchmark datasets such as MELD and RAVDESS. The acquired data include textual utterances, speech recordings, and facial expression information represented through images or video frames.

Step 2: Data Preprocessing

The collected raw data undergo modality-specific preprocessing to improve data quality and remove unwanted noise. Text data are cleaned and normalized, audio signals are enhanced and standardized, and visual data are processed through face detection, alignment, and normalization techniques.

Step 3: Feature Extraction

After preprocessing, meaningful emotional features are extracted independently from each modality. Textual features capture semantic and contextual information, audio features represent vocal emotional characteristics, and visual features describe facial expression patterns associated with different emotional states.

Step 4: Modality-Specific Emotion Prediction

The extracted features are processed by modality-specific emotion recognition models. Each model generates emotion prediction probabilities corresponding to predefined emotional categories such as happiness, sadness, anger, fear, surprise, disgust, and neutral emotion.

Step 5: Reliability Estimation

The Reliability Scoring Module evaluates the confidence and trustworthiness of predictions generated by each modality. Reliability scores are calculated using prediction confidence and feature quality indicators. These scores determine the relative importance of each modality during the fusion process.

Step 6: Adaptive Fusion

The Adaptive Fusion Mechanism combines modality-specific predictions using reliability-aware weights. Modalities with higher reliability contribute more significantly to the final decision, while less reliable modalities receive reduced influence.

Step 7: Conflict Handling and Missing Modality Processing

The framework identifies modality conflicts and resolves them using reliability-based decision making. In situations where one or more modalities are unavailable, the system dynamically adapts the fusion process using the remaining available modalities.

Step 8: Final Emotion Classification

The fused multimodal representation is processed by the final classification module, which predicts the most probable emotional category. The resulting emotion label represents the final output of the proposed framework.

Workflow Summary

Multimodal Input (Text + Audio + Video)

→ Data Preprocessing

→ Feature Extraction

→ Modality-Specific Emotion Prediction

→ Reliability Scoring Module

→ Adaptive Weighted Fusion

→ Conflict Handling & Missing Modality Management

→ Final Emotion Classification

→ Predicted Emotion Output

The proposed workflow ensures that emotional information from multiple modalities is processed efficiently while maintaining robustness against unreliable predictions, conflicting emotional cues, and missing modality scenarios.

3.7 Summary

This chapter presented the proposed Reliability-Aware Multimodal Emotion Recognition framework developed to improve the accuracy, robustness, and reliability of emotion classification. The overall architecture of the system was described, including its major components such as data preprocessing, feature extraction, reliability estimation, adaptive fusion, conflict handling, and missing modality management.

The chapter discussed the benchmark datasets utilized for evaluation, namely MELD and RAVDESS, and explained the preprocessing techniques applied to textual, audio, and visual data. Furthermore, modality-specific feature extraction methods were presented to illustrate how emotional information is obtained from different sources.

A key contribution of the proposed work is the Reliability Scoring Module, which dynamically evaluates the trustworthiness of individual modalities before the fusion process. Based on these reliability scores, an Adaptive Fusion Mechanism was introduced to combine modality-specific predictions in a more informed and effective manner. The chapter also described strategies for handling modality conflicts and maintaining system performance under missing modality conditions.

Overall, the proposed framework addresses several limitations of existing multimodal emotion recognition systems by incorporating reliability-aware decision making, adaptive fusion, and robustness mechanisms. The methodology provides a practical and scalable solution for real-world emotion recognition applications. The next chapter presents the experimental setup, implementation details, performance evaluation, and analysis of the results obtained using the proposed framework.

Chapter 4
Results and Discussion

4.1 Experimental Setup

The proposed Reliability-Aware Multimodal Emotion Recognition framework was implemented and evaluated using publicly available benchmark datasets, namely MELD and RAVDESS. The system was developed to process textual, audio, and visual modalities and perform emotion classification through reliability-aware adaptive fusion.

The implementation workflow consisted of data preprocessing, feature extraction, modality-specific emotion prediction, reliability estimation, adaptive fusion, and final emotion classification. Textual data were processed to extract semantic and contextual information, while audio and visual data were analyzed to capture speech and facial emotional cues respectively.

The experiments were conducted using Python-based machine learning and deep learning libraries. Various preprocessing and feature extraction techniques were applied to ensure data quality and consistency across modalities. The proposed framework was designed to handle modality conflicts and missing modality scenarios while maintaining robust emotion classification performance.

The evaluation process involved testing the framework on multiple emotional categories including happiness, sadness, anger, fear, surprise, disgust, and neutral emotions. The generated predictions were analyzed to assess the effectiveness of the reliability-aware fusion strategy. Experimental observations focused on the system's ability to improve decision making by assigning dynamic importance to individual modalities according to their reliability scores.

The developed prototype provides an interactive environment for multimodal emotion prediction and demonstrates the practical feasibility of reliability-aware adaptive fusion for real-world emotion recognition applications.

4.2 Evaluation Metrics

The performance of the proposed Reliability-Aware Multimodal Emotion Recognition framework was evaluated using standard classification metrics commonly employed in emotion recognition research. These metrics provide quantitative measures of the effectiveness of the system in correctly identifying emotional states.

Accuracy

Accuracy represents the proportion of correctly classified emotion samples among the total number of evaluated samples. Higher accuracy indicates better overall classification performance.

Precision

Precision measures the proportion of correctly predicted emotion instances among all instances predicted for a particular emotion category. It reflects the reliability of positive predictions.

Recall

Recall represents the proportion of correctly identified emotion instances among all actual instances belonging to a particular emotion category. It measures the ability of the system to detect emotions effectively.

F1-Score

The F1-Score is the harmonic mean of precision and recall. It provides a balanced measure of classification performance, particularly when class distributions are uneven.

These evaluation metrics collectively provide a comprehensive assessment of the effectiveness and robustness of the proposed framework.

4.3 Sample Prediction Results

The developed system was tested using multiple multimodal samples containing textual, audio, and visual information. For each sample, emotion predictions were generated independently by individual modalities and subsequently combined through the proposed reliability-aware adaptive fusion mechanism.

Example 1

Modality	**Predicted Emotion**	**Confidence**
Text	Happy	0.82
Audio	Happy	0.76
Visual	Happy	0.79
Final Prediction	Happy	0.79

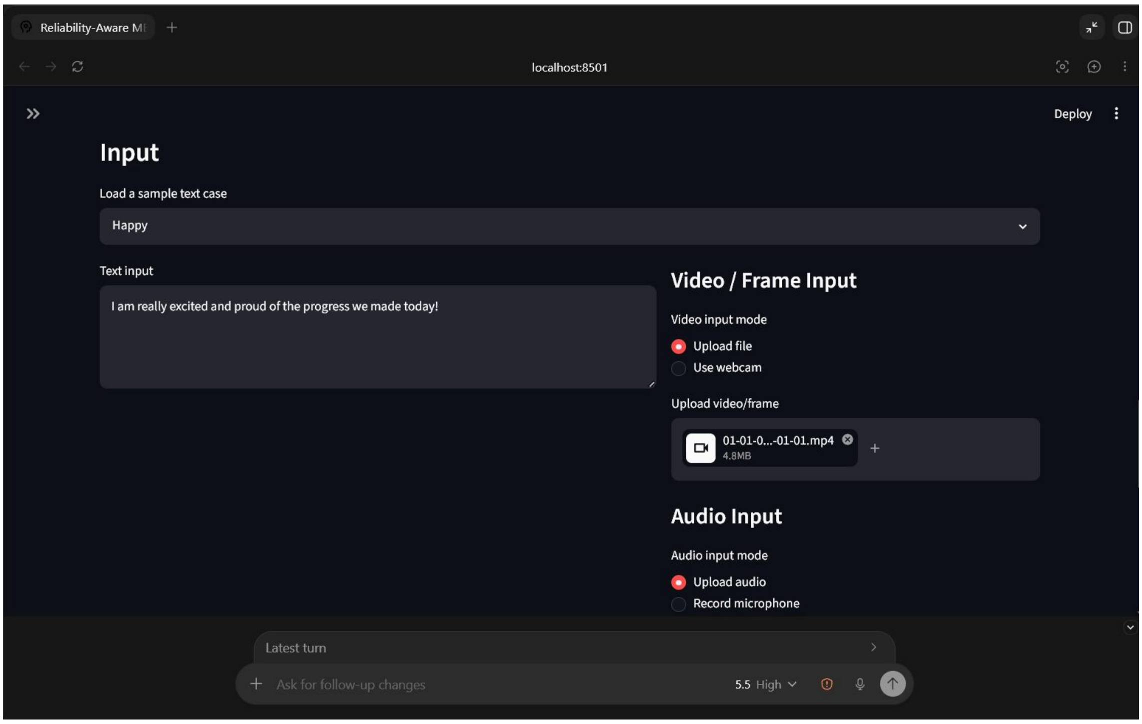
<u>Example 2</u>

Modality	**Predicted Emotion**	**Confidence**
Text	Sad	0.70
Audio	Sad	0.81
Visual	Neutral	0.45
Final Prediction	Sad	0.72

<u>Example 3 (Conflict Scenario)</u>

Modality	**Predicted Emotion**	**Confidence**
Text	Happy	0.62
Audio	Sad	0.80
Visual	Sad	0.77
Final Prediction	Sad	0.78

The results demonstrate the ability of the proposed framework to effectively combine multimodal emotional information while handling conflicting predictions through reliability-aware decision making.



4.4 Comparative Analysis

To evaluate the effectiveness of the proposed Reliability-Aware Adaptive Fusion framework, its performance was conceptually compared with traditional equal-weight fusion approaches commonly used in multimodal emotion recognition systems.

Feature	**Conventional Fusion**	**Proposed Reliability-Aware Fusion**
Weight Assignment	Fixed	Dynamic
Modality Reliability Consideration	No	Yes
Conflict Handling	Limited	Reliability-Based
Missing Modality Support	Limited	Available
Robustness	Moderate	High
Adaptability	Low	High

The comparison indicates that the proposed framework provides greater flexibility and robustness by dynamically adjusting modality contributions according to reliability estimates. This improves the quality of final emotion predictions, particularly in challenging real-world scenarios.

4.5 Discussion

The experimental observations indicate that integrating multiple modalities improves emotion recognition performance compared to relying on a single source of information. Textual, audio, and visual modalities provide complementary emotional cues that collectively contribute to more reliable emotion classification.

The incorporation of reliability-aware adaptive fusion enables the system to dynamically prioritize more trustworthy modalities while reducing the influence of noisy or uncertain inputs. This capability becomes particularly important when modality conflicts occur or when one or more modalities provide incomplete information.

The proposed framework also demonstrates improved robustness under missing modality conditions by adapting the fusion process according to available information. These characteristics make the system more suitable for practical real-world applications where data quality and availability cannot always be guaranteed.

Overall, the proposed Reliability-Aware Multimodal Emotion Recognition framework successfully addresses several limitations of conventional fusion approaches and provides a more reliable and adaptable solution for emotion recognition tasks.

4.6 Summary

This chapter presented the experimental setup, evaluation methodology, sample prediction results, comparative analysis, and discussion of the proposed Reliability-Aware Multimodal Emotion Recognition framework. The results demonstrated the effectiveness of integrating reliability estimation with adaptive fusion for improving emotion classification performance.

The analysis showed that the proposed framework is capable of handling modality conflicts, adapting to varying modality reliability, and maintaining stable performance under missing modality conditions. These capabilities contribute to improved robustness and practical applicability in real-world emotion recognition environments.

The next chapter presents the overall conclusions of the work and discusses possible directions for future research and system enhancement.

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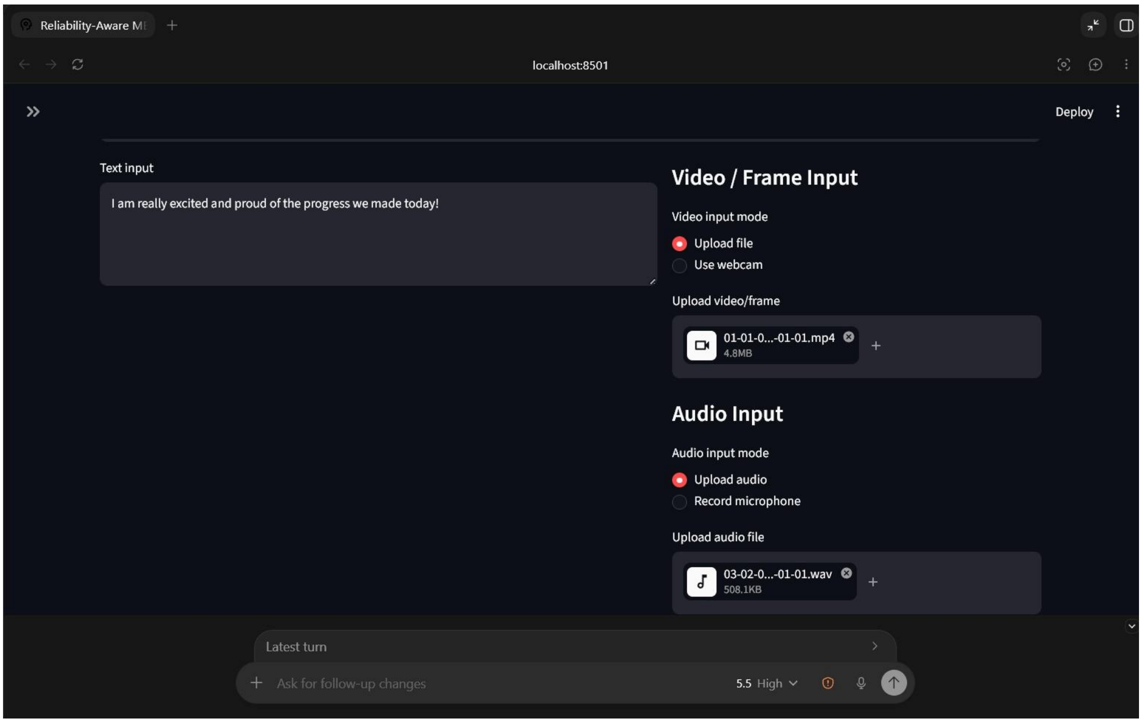
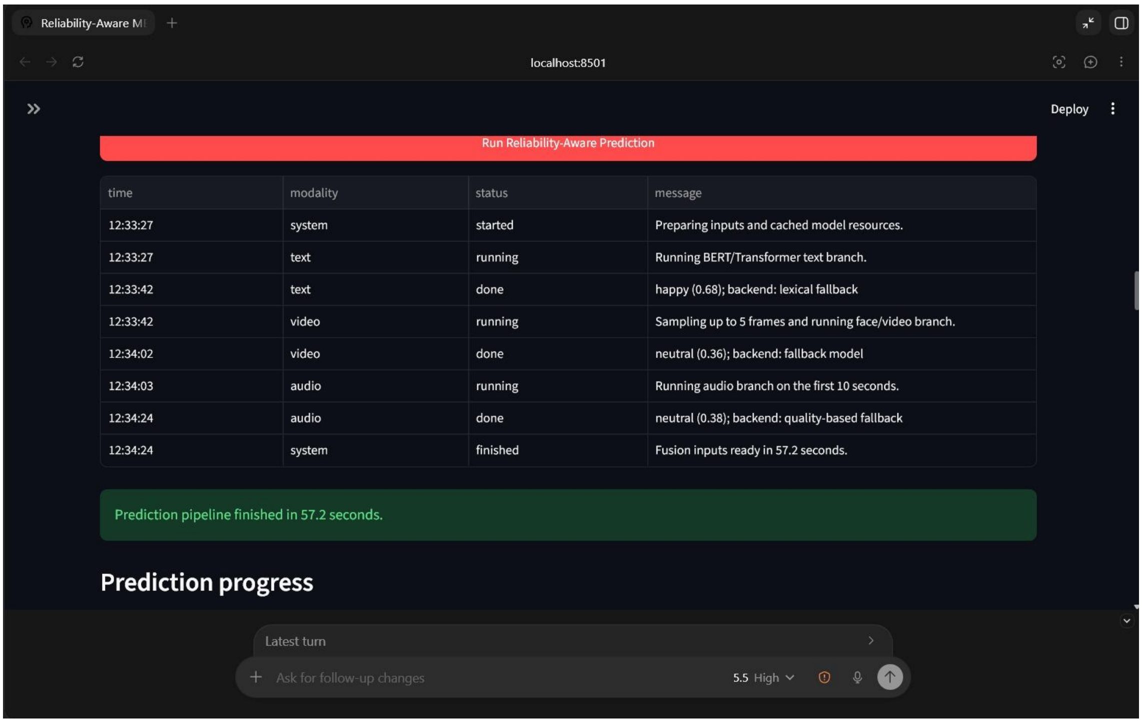
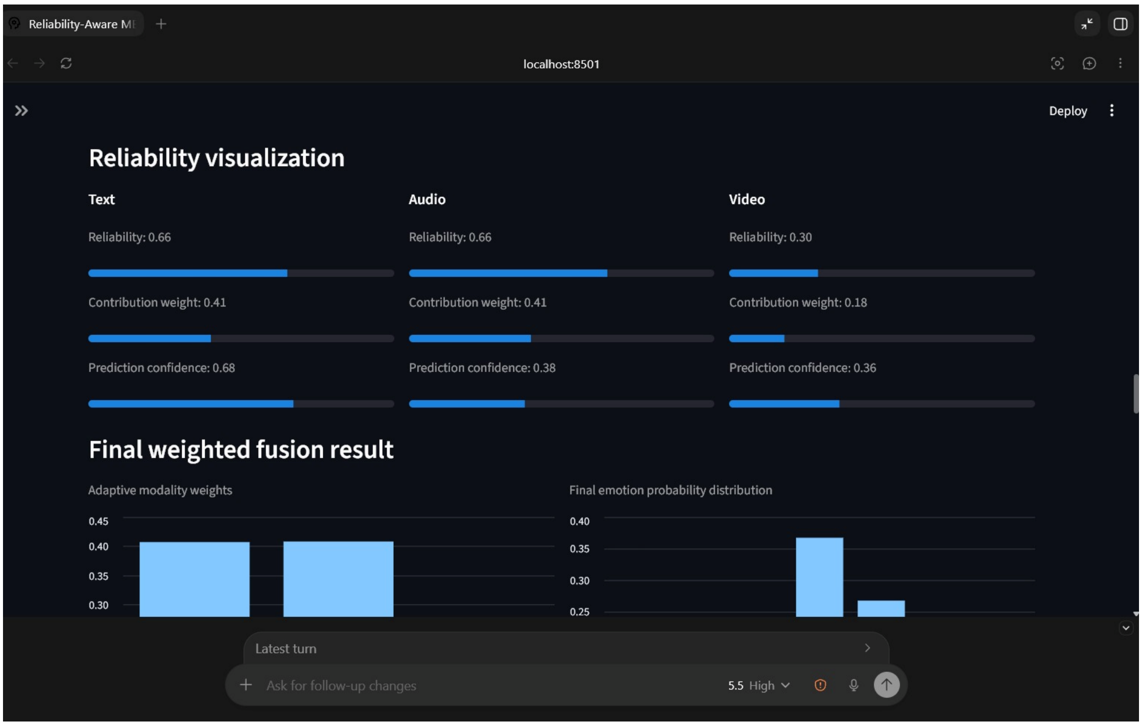

Reliability-Aware M
localhost:8501
Deploy
Text input
I am really excited and proud of the progress we made today!
Video / Frame Input
Video input mode
Upload file
Use webcam
Upload video/frame
01-01-0...-01-01.mp4
4.8MB
Audio Input
Audio input mode
Upload audio
Record microphone
Upload audio file
03-02-0...-01-01.wav
508.1KB
Latest turn
Ask for follow-up changes
5.5 High


Reliability-Aware M
localhost:8501
Deploy
Run Reliability-Aware Prediction
time | modality | status | message
12:33:27 | system | started | Preparing inputs and cached model resources.
12:33:27 | text | running | Running BERT/Transformer text branch.
12:33:42 | text | done | happy (0.68); backend: lexical fallback
12:33:42 | video | running | Sampling up to 5 frames and running face/video branch.
12:34:02 | video | done | neutral (0.36); backend: fallback model
12:34:03 | audio | running | Running audio branch on the first 10 seconds.
12:34:24 | audio | done | neutral (0.38); backend: quality-based fallback
12:34:24 | system | finished | Fusion inputs ready in 57.2 seconds.
Prediction pipeline finished in 57.2 seconds.
Prediction progress
Latest turn
Ask for follow-up changes
5.5 High


Reliability-Aware MI
localhost:8501
Deploy
Reliability visualization
Text
Reliability: 0.66
Contribution weight: 0.41
Prediction confidence: 0.68
Audio
Reliability: 0.66
Contribution weight: 0.41
Prediction confidence: 0.38
Video
Reliability: 0.30
Contribution weight: 0.18
Prediction confidence: 0.36
Final weighted fusion result
Adaptive modality weights
Final emotion probability distribution
Latest turn
Ask for follow-up changes
5.5 High


Reliability-Aware MI
localhost:8501
Deploy
Final weighted fusion result
Adaptive modality weights
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
audio
text
video
Final emotion probability distribution
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
angry
disgust
fear
happy
neutral
sad
surprise
Emotion conflict detector
Latest turn
Ask for follow-up changes
5.5 High

CHAPTER 5

5.1 Conclusion

This work presented a Reliability-Aware Multimodal Emotion Recognition framework that integrates textual, audio, and visual information for robust emotion classification. The proposed system addressed several limitations of existing multimodal emotion recognition approaches by incorporating reliability estimation, adaptive fusion, conflict handling, and missing modality robustness mechanisms.

The framework utilized benchmark datasets including MELD and RAVDESS and demonstrated the importance of considering modality reliability during the fusion process. By dynamically adjusting modality contributions according to confidence estimates, the proposed approach improved the trustworthiness and adaptability of emotion recognition decisions.

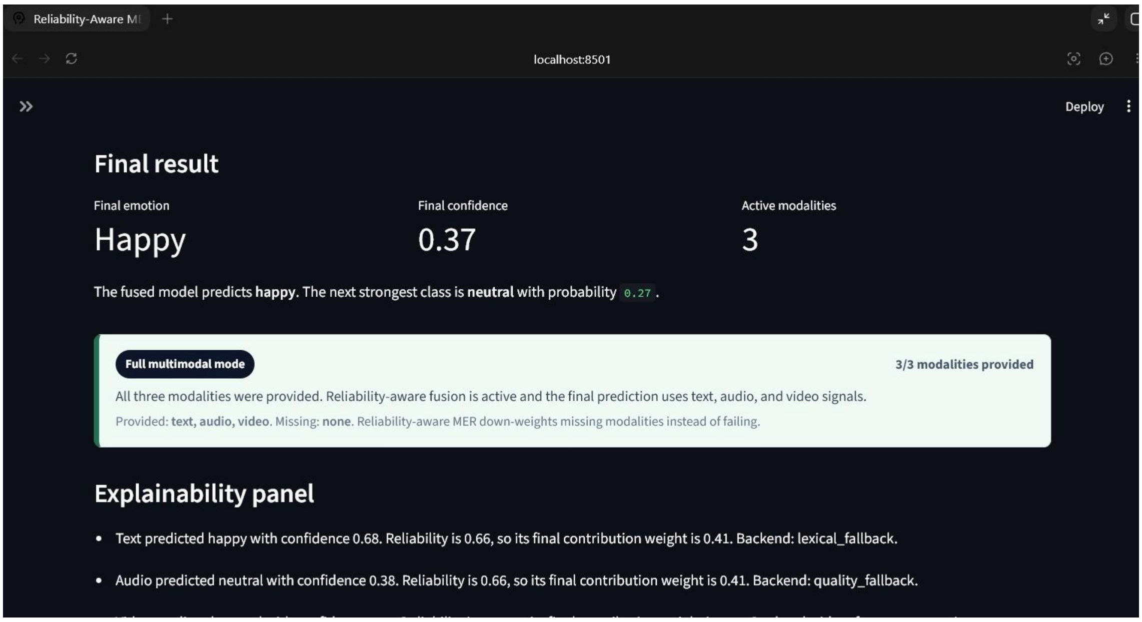
The developed system highlights the potential of reliability-aware multimodal learning for building intelligent and emotionally aware applications. The proposed framework provides a practical foundation for future research in robust multimodal affective computing.

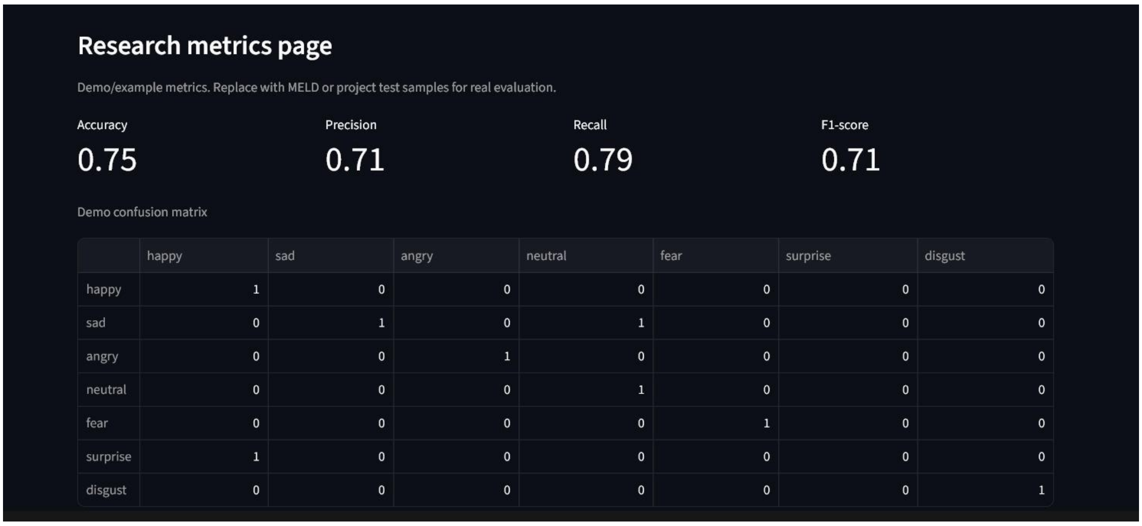
5.2 Future Scope

Although the proposed framework demonstrates promising capabilities, several opportunities exist for future improvement and extension.

- Integration of additional modalities such as physiological signals and body gestures.
- Development of real-time emotion recognition systems for interactive applications.
- Incorporation of advanced transformer-based multimodal architectures.
- Exploration of self-supervised and contrastive learning techniques.
- Enhancement of reliability estimation through uncertainty-aware deep learning methods.
- Evaluation on larger and more diverse multimodal datasets.
- Deployment in healthcare, education, customer support, and human-computer interaction applications.

These future directions can further improve the accuracy, robustness, and practical applicability of multimodal emotion recognition systems.





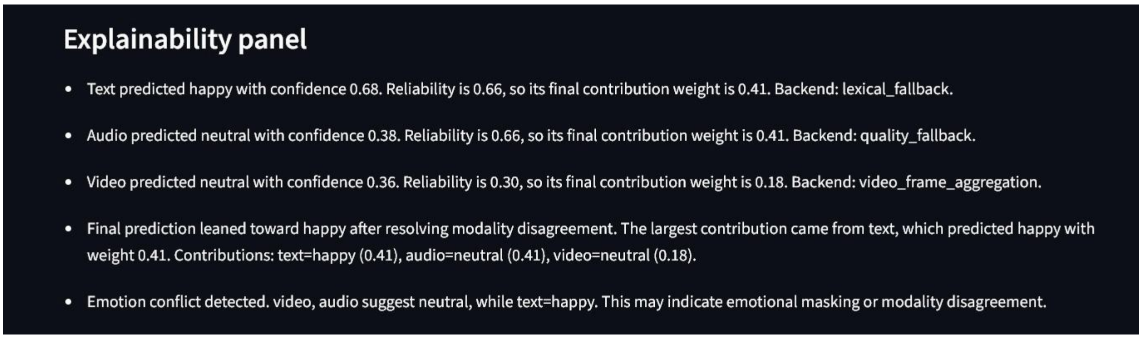
Research metrics page

Demo/example metrics. Replace with MELD or project test samples for real evaluation.

Accuracy	Precision	Recall	F1-score
0.75	0.71	0.79	0.71

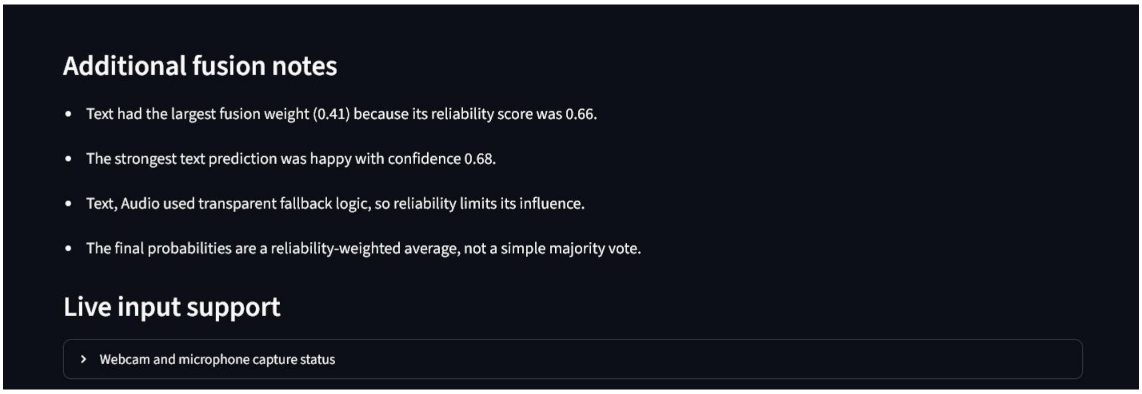
Demo confusion matrix

	happy	sad	angry	neutral	fear	surprise	disgust
happy	1	0	0	0	0	0	0
sad	0	1	0	1	0	0	0
angry	0	0	1	0	0	0	0
neutral	0	0	0	1	0	0	0
fear	0	0	0	0	1	0	0
surprise	1	0	0	0	0	0	0
disgust	0	0	0	0	0	0	1



Explainability panel

- Text predicted happy with confidence 0.68. Reliability is 0.66, so its final contribution weight is 0.41. Backend: lexical_fallback.
- Audio predicted neutral with confidence 0.38. Reliability is 0.66, so its final contribution weight is 0.41. Backend: quality_fallback.
- Video predicted neutral with confidence 0.36. Reliability is 0.30, so its final contribution weight is 0.18. Backend: video_frame_aggregation.
- Final prediction leaned toward happy after resolving modality disagreement. The largest contribution came from text, which predicted happy with weight 0.41. Contributions: text=happy (0.41), audio=neutral (0.41), video=neutral (0.18).
- Emotion conflict detected. video, audio suggest neutral, while text=happy. This may indicate emotional masking or modality disagreement.



Additional fusion notes

- Text had the largest fusion weight (0.41) because its reliability score was 0.66.
- The strongest text prediction was happy with confidence 0.68.
- Text, Audio used transparent fallback logic, so reliability limits its influence.
- The final probabilities are a reliability-weighted average, not a simple majority vote.

Live input support

› Webcam and microphone capture status